BANK CUSTOMER INSIGHT

DS Sprint Skill-Based Assessment 2 | Use Case 08 | TrustWave Bank

Synthetic data generated with fixed seed 20261013. Findings and model metrics are illustrative and do not describe real bank customers.

Student Details

Name	Vissvapriya P	Register Number	_____
Class	III CSE D	Challenge ID	CH08
Challenge Title	Bank Customer Insight	Sector	Banking

Round 1 - Data Understanding, Preparation & EDA

1. Problem Statement

This project analyzes TrustWave Bank's synthetic customer, account, transaction, channel, and service data to understand engagement, channel preferences, account activity, and complaints. The aim is to identify measurable patterns and recommend practical service improvements.

2. Dataset Details

All five source tables are separate linked CSV datasets; counts below are verified directly from the raw CSV files.

Dataset	Records	Attributes	Actual fields
customers.csv	1,200	4	customer_id, age_band, region, relationship_months
accounts.csv	1,907	4	account_id, customer_id, account_type, balance
transactions.csv	48,169	5	transaction_id, transaction_date, amount, transaction_type, account_id
channels.csv	1,200	5	customer_id, branch_uses, atm_uses, mobile_uses, internet_uses
service.csv	1,971	7	service_id, customer_id, request_date, request_type, is_complaint, resolution_status, resolution_days

These separate tables total 54,447 source rows (not distinct customers). The merged customer feature table is 1,200 rows by 14 columns; the target is a separate table.

3. Data Preprocessing

Filled 277 missing channel values using age-band medians. Standardized account labels. Removed 476 duplicate transaction IDs (48,169 raw rows to 47,693 cleaned rows), retaining other uniquely identified similar transactions.

Used IQR amount fences of $0.00 and $279.81; flagged and capped 3,655 values while retaining the original amount. Filled 370 missing service cells with explicit labels or the observed median. Joined tables on customer/account IDs and aggregated behavioral counts to customer level.

4. Two EDA Findings

1. **Mobile banking was the most-used channel**, with 5,286 reported uses.

2. **North had the highest complaint count**, with 169 complaint records.

Round 2 - Feature Engineering & Machine Learning

1. Selected Features

Implemented inputs: Age band, Region, Relationship length (months), Total account balance, Account count, Transaction count, Complaint count, Service-request count, Branch uses, ATM uses, Mobile uses, Internet uses, Transactions per month.

Target: customer engagement segment (Low, Medium, High), defined by tertiles of a simulated latent engagement propensity. The latent value itself is excluded from model input to avoid direct target leakage.

2. ML Algorithm Used

Compared DummyClassifier baseline, Logistic Regression, and Random Forest Classifier. The selected model is **Logistic Regression**, based on the highest test macro F1-score (accuracy breaks ties).

3. Justification

Logistic Regression is a suitable, comparatively interpretable baseline for three-class classification. Standardization for numeric inputs and one-hot encoding for categorical inputs are fitted inside Scikit-learn pipelines, using a shared stratified 75/25 split for a fair comparison.

4. Model Performance

Holdout test set: 300 customers; training set: 900. Selected model test accuracy: 0.7033 (70.33%); macro F1-score: 0.6991.

Model	Test accuracy	Test macro F1
DummyClassifier	0.3333	0.1667
Logistic Regression	0.7033	0.6991
Random Forest	0.6967	0.6945

Class	Precision	Recall	F1-score	Support
Low	0.7414	0.8600	0.7963	100
Medium	0.5714	0.5200	0.5445	100
High	0.7849	0.7300	0.7565	100

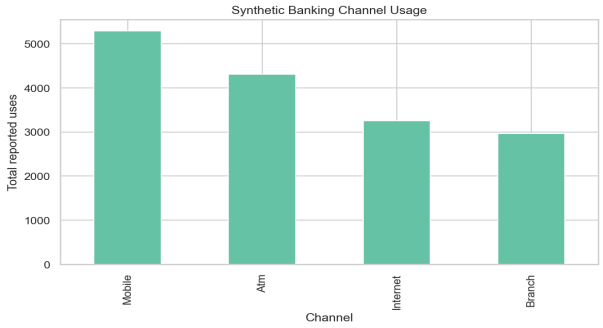
Interpretation: the selected classifier correctly labeled 70.33% of this synthetic test set; macro F1 summarizes balanced class performance and is 0.6991. These are prototype results, not validated production estimates.

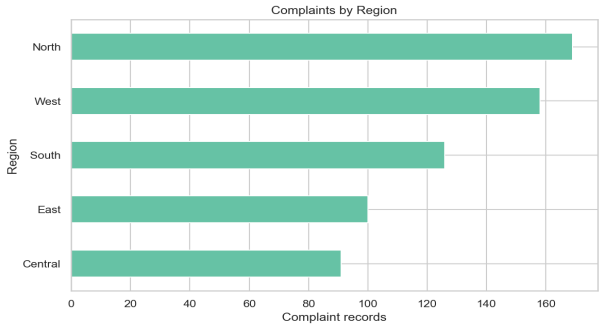
Round 3 - Intelligence Layer, Visualization & Prototype

1. Visualization

Generated chart	Saved file
Account activity, Q1 vs Q4	account_activity.png
Average channel usage by age band	age_group_comparison.png
Balance by engagement segment	balance_distribution.png
Banking-channel usage	channel_usage.png
Complaint counts by region	regional_complaints.png
Service requests by type	service_requests.png
Monthly transaction trend	transaction_trends.png

The Excel workbook contains a Summary Dashboard, 9 analysis/data sheets, 9 formatted tables, and 2 embedded charts. The implemented Streamlit app provides age-band and region filters and shows engagement, channels, account activity, complaints, and service trends.





2. Key Insights

Mobile led channel usage (5,286 uses), and North had the most complaints (169).

Term Deposit activity declined 20.99% from Q1 (3,620) to Q4 (2,860) 2025. Card support and Payments tied for most repeated request types at 60 customer/service-type pairs each.

3. Prototype / Intelligence Layer

Python data preparation and Scikit-learn pipelines create and evaluate customer engagement classifications. The working Streamlit application presents filtered customer, channel, account, complaint, and service insights. The Excel workbook is a separate analysis deliverable; the PowerPoint pitch is in the presentation folder.

4. Recommendations

1. Pilot assisted digital onboarding for customers aged 60+ (average mobile 2.16, internet 1.31 uses, the lowest combined age-band average); measure adoption before scaling.

2. Review complaint drivers and resolution in North (169 complaints), investigate Card support and Payments repeat contacts (60 each), and test targeted Term Deposit follow-up after the measured activity decline.

Excel Analysis, Business Pitch & Final Submission

Round 3 - Data Analysis Using Excel: Evidence Checklist

Workbook inspected: `excel/bank_customer_insight_dashboard.xlsx`. Status describes the workbook itself, not Python charts or reports.

Assessment item	Status	Verified workbook evidence
Cleaned data imported into Excel	Complete	Cleaned customer, account, transaction and service-analysis data are included in workbook sheets.
Data organization and formatting	Complete	10 sheets; 9 formatted tables; styled headers and frozen panes.
Formulas and statistical functions	Partial	2 Excel formulas (COUNTA); no Excel statistical formulas. Several summaries are precomputed in Python.
Channel usage analysis	Complete	Channel Analysis sheet has totals for all four channels.
Customer segment comparison	Complete	Customer Segments and Balance Analysis sheets include engagement classes and segment balances.
Account activity analysis	Complete	Account Activity sheet compares Q1 and Q4 transactions by account type.
Complaint analysis	Complete	Complaint Analysis sheet reports complaint counts by region.
Balance distribution analysis	Complete	Balance Analysis sheet reports balance statistics by segment.
Service-request trend analysis	Complete	Service Trends sheet contains request and complaint counts by request type.
Charts	Partial	2 Excel charts are embedded (channel usage and account activity); other analyses are tables only.
PivotTables and PivotCharts	Missing	Workbook contains 0 native PivotTables/PivotCharts; it includes instructions for creating PivotTables in Excel.
Summary sheet with findings	Complete	Summary Dashboard presents findings, formulas, model results, charts, and PivotTable instructions.

PivotTables/PivotCharts were not generated natively. The workbook includes desktop Excel PivotTable instructions; create these manually if the assessment requires native pivots.

Round 4 - Pitching Business Insights

TrustWave Bank needs to understand how customers use its channels and where service friction may be reducing engagement. In this synthetic sample, mobile banking led with 5,286 reported uses, while North recorded the largest complaint count at 169. These patterns make digital access and regional service quality useful areas for management review, but counts should be validated against real customer volumes. Term Deposit activity was 20.99% lower in Q4 than Q1, a reason to investigate whether product journeys or seasonal behavior merit follow-up. Repeat requests for Card support and Payments were tied at 60 customer-service-type pairs each. We recommend piloting assisted digital onboarding for older customers, reviewing North complaint causes and repeat-request handling, and testing a Term Deposit retention contact. Measure adoption, complaint resolution, and repeat contacts before scaling any intervention.

Pitch length: 131 words.

Final Submission

GitHub Repository Link: _____

Overall Result / Conclusion: The synthetic-data workflow links customer, channel, account, transaction, and service patterns and compares three engagement classifiers. Mobile led channel use, North had the highest complaint count, and Logistic Regression achieved 70.33% test accuracy with macro F1 0.6991. Validate insights against governed real data before operational use.